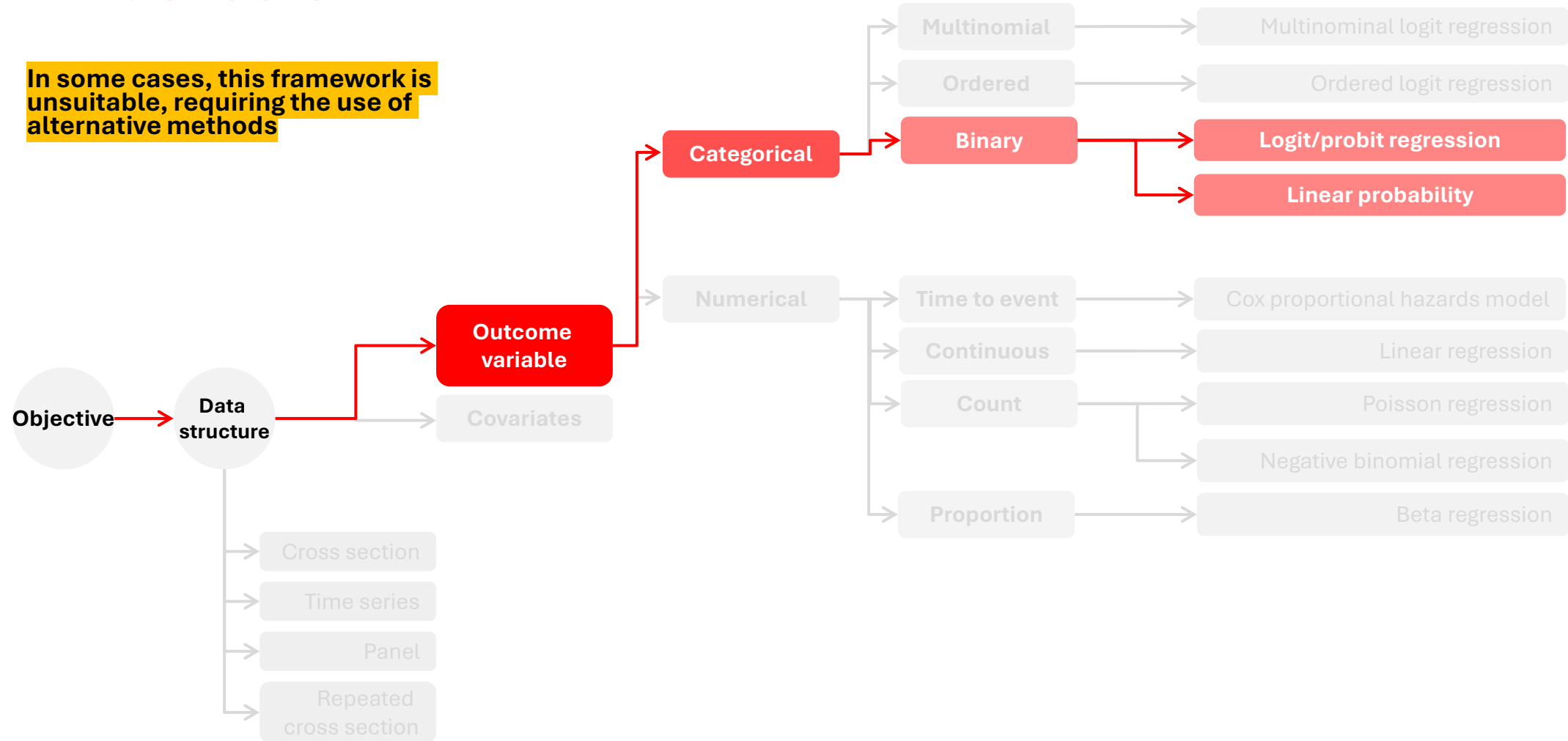


Binomial models

Data types and model selection

In some cases, this framework is unsuitable, requiring the use of alternative methods



Binomial Models

A binomial response variable can take only two mutually exclusive values, typically coded as 0 and 1.

$$\text{Familias en acción}(y) = \begin{cases} 1 & \text{if } y = \text{in} \\ 0 & \text{if } y = \text{out} \end{cases}$$

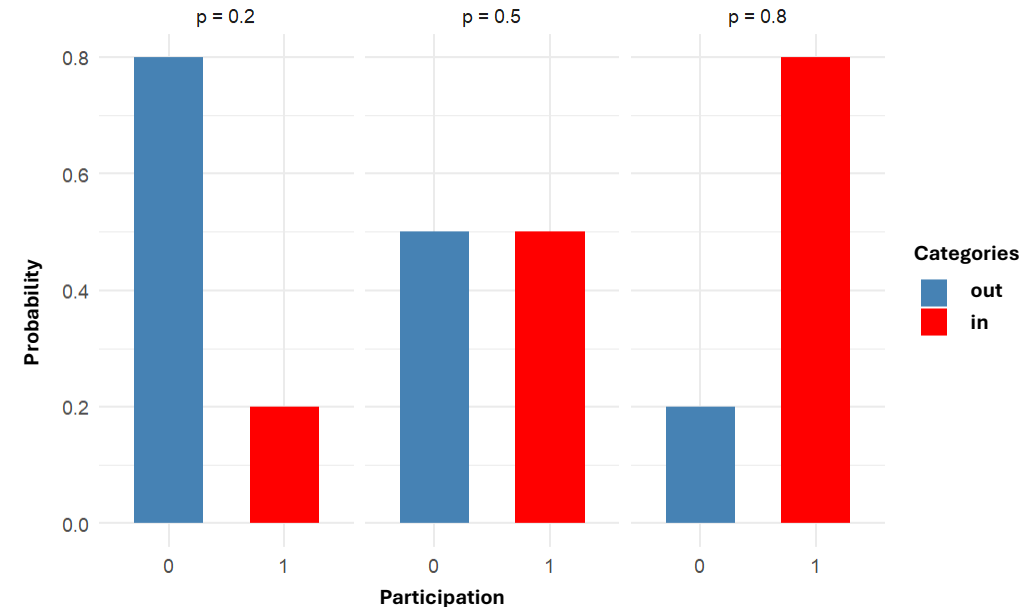
For such variables, the relevant statistical distribution is a **binomial distribution with a single trial**.

$$P(Y = y) = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{if } y = 0 \end{cases}$$

Therefore, in this context, one should not assume the distribution of the response variable; rather, the **binomial framework inherently defines it**.

Bernoulli distribution

Familias en acción



Croissant, Y. (2024). *Microeconometrics with R* (1st ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/9781003100263>
 Cameron, A. C., & Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications* (13th printing). Cambridge University Press.
 Logit Regression | R Data Analysis Examples. (n.d.). Retrieved September 28, 2025, from <https://stats.oarc.ucla.edu/r/dae/logit-regression/>
 Getting Started in Logit and Ordered Logit Regression. (n.d.). Oscar Torres-Reyna. <http://www.princeton.edu/~otorres/>

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Therefore, in this context, one should not assume the distribution of the response variable; rather, the **binomial framework inherently defines it**.

If we define the two possible outcomes of a random variable as:

- 1 as success
- 0 as failure

Then the distribution is fully characterised by a single parameter, μ , **which represents the probability of success**, p , and is also the expected value of the variable.

$$\begin{aligned} \mathbb{E}(Y) &= (1 - p) \cdot 0 + p \cdot 1 \\ \mathbb{E}(Y) &= (1 - p) \cdot 0 + p \cdot 1 = \mu \\ \mathbb{E}(Y) &= p = \mu \end{aligned}$$

The variance of the binomial response is given by:

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 \\ \text{Var}(Y) &= \mathbb{E}(Y) - \mu^2 \\ \text{Var}(Y) &= \mu - \mu^2 \\ \text{Var}(Y) &= \mu \cdot (1 - \mu) \end{aligned}$$

Binomial Models

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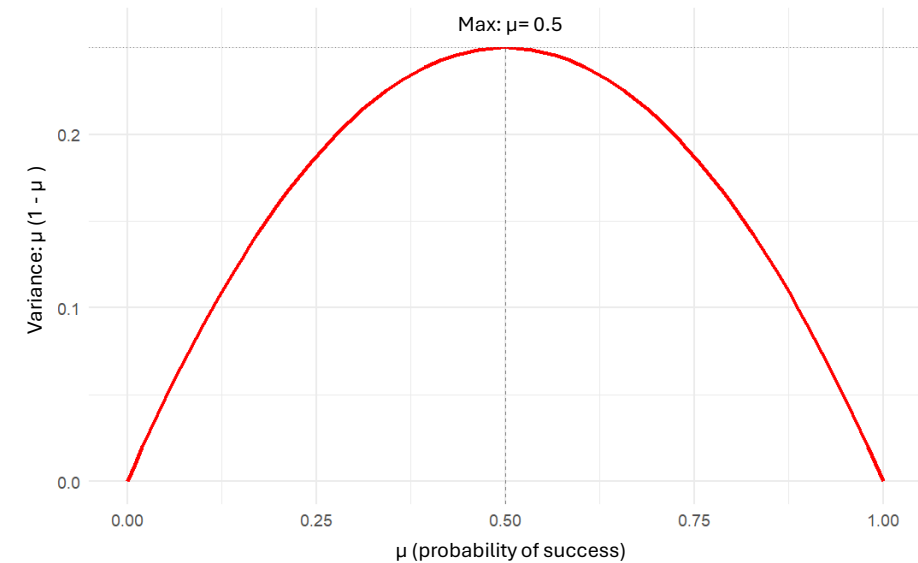
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Variance of a Bernoulli distribution



As the mean of the variable approaches 1 or 0, the variance of Y tends towards zero.

- This occurs because, in such extreme cases, almost all observed values in the sample will be either 1 or 0, leaving very little variability.

Binomial Models

Index function

To construct a regression model for a binomial response, we first define an index function, **also known as the linear predictor**:

$$\eta_n = \gamma^\top z_n = \alpha + \beta^\top x_n$$

This index function is then linked to the **single parameter of the binomial distribution**, namely the probability of success μ_n , through a chosen **link function F** .

$$\mu_n = F(\eta_n)$$

Different choices of the function F lead to different binomial models. After defining the index function, we select a function F that maps the linear predictor to the binomial probability.

1. Linear Probability Model

Uses the identity function as the link. Simple to estimate via OLS, but may produce probabilities outside the $[0,1]$ interval.

2. Logit model

Uses the logistic cumulative distribution function as the link. Ensures probabilities remain between 0 and 1.

3. Probit model

Uses the standard normal cumulative distribution function as the link. Also ensures probabilities remain between 0 and 1.

Binomial Models

Linear probability model

The most direct choice for the link function F is the identity function:

$$\mu_n = \eta_n = \alpha + \beta^\top x_n$$

This implies that the model can be estimated using the familiar Ordinary Least Squares (OLS) approach, **given its linear structure**. Since the model is linear:

$$\frac{\partial \mu_n}{\partial x_{kn}} = \beta_k$$

- The parameters of the binomial distribution are assumed to be a linear function of the covariates x_n .
- The estimated **coefficients can be directly interpreted as constant marginal effects of each covariate** on the probability of success.

⊘ Limitations

1. Residual Structure

Because the outcome y takes only the values 0 or 1, **the residual is mechanically related to the covariates**:

$$r_n = y_n - \hat{\mu}_n = \begin{cases} -(\hat{\alpha} + \hat{\beta}x_n) & \text{if } y_n = 0 \\ 1 - (\hat{\alpha} + \hat{\beta}x_n) & \text{if } y_n = 1 \end{cases}$$

This creates correlation between the error term and the regressors, **violating the OLS assumption of independence** and affecting the consistency of the estimators. Additionally, the **residual variance is heteroskedastic**, which means:

- OLS becomes inefficient
- the reported standard errors are biased, unless corrected.

Binomial Models

Linear probability model

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⊘ Limitations

2. Predicted probabilities may fall outside [0,1]

Since the identity function does not constrain predictions, the model may produce:

- Probabilities below 0
- Probabilities above 1

Which are not meaningful for a probability model. Therefore, we typically prefer a link function F that ensures all predicted values lie within the $[0,1]$ interval.

- **Monotonicity:** as the linear predictor $z = \alpha + \beta^\top x$ increases, the value of $F(z)$ should also increase. In probabilistic terms, higher values of the covariates should correspond to a higher probability of success.
- **Correct limits:** When z becomes very negative, the predicted probability should approach 0 but never become negative.
- **Upper bound:** When z becomes very large, the predicted probability should approach 1 but never exceed it.

Binomial Models

Motivation for Non-Linear Models

These issues motivate the use of alternative link functions which form the basis of logit and probit models. **These models guarantee that predicted probabilities always remain in the valid range and are therefore more appropriate for binary outcomes.**

Normal function

$$\Phi(z) = \int_{-\infty}^z \phi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

The probability that a standard normal random variable takes a value less than or equal to z .

The probability that a standard normal random variable takes a value less than or equal to z .

$$\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

Indicates that we are accumulating the area under the density curve from minus infinity up to z .

Logistic function

$$\Lambda(z) = \frac{e^z}{1 + e^z}$$

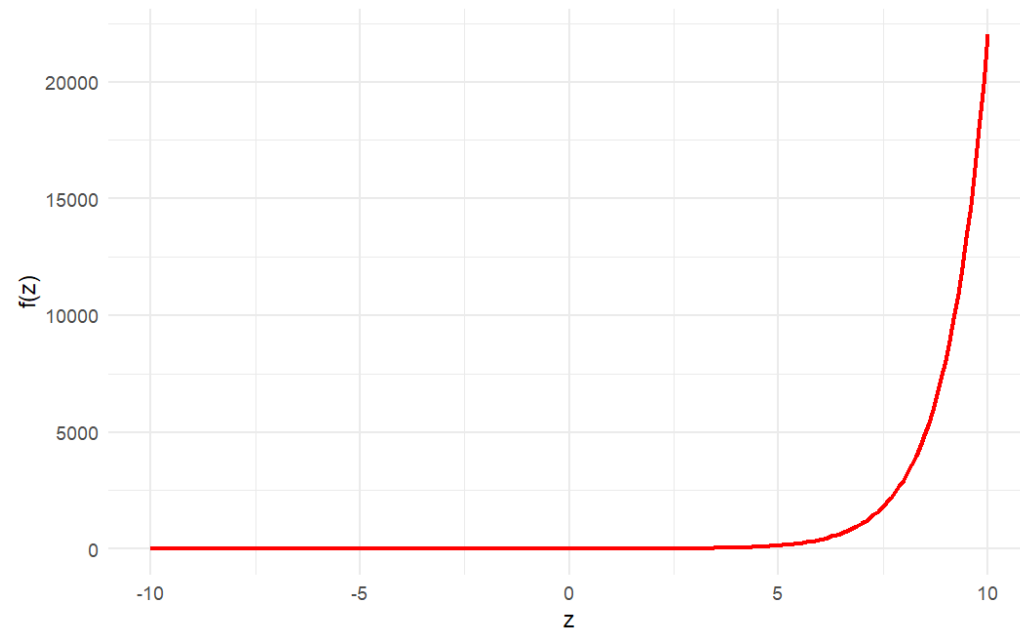
The denominator that normalises the exponential growth, ensuring that the result lies between 0 and 1.

It is the exponential function applied to the linear predictor z . The denominator grows rapidly as z increases and tends towards zero as z decreases.

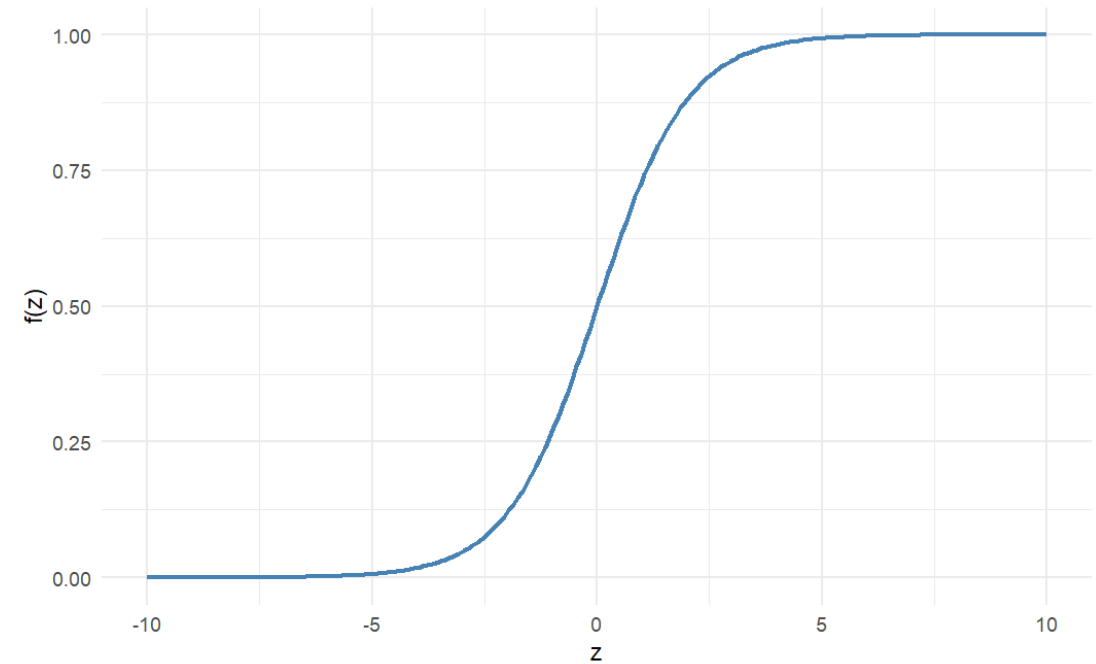
Binomial Models

Logistic function

$$g(z) = e^z$$



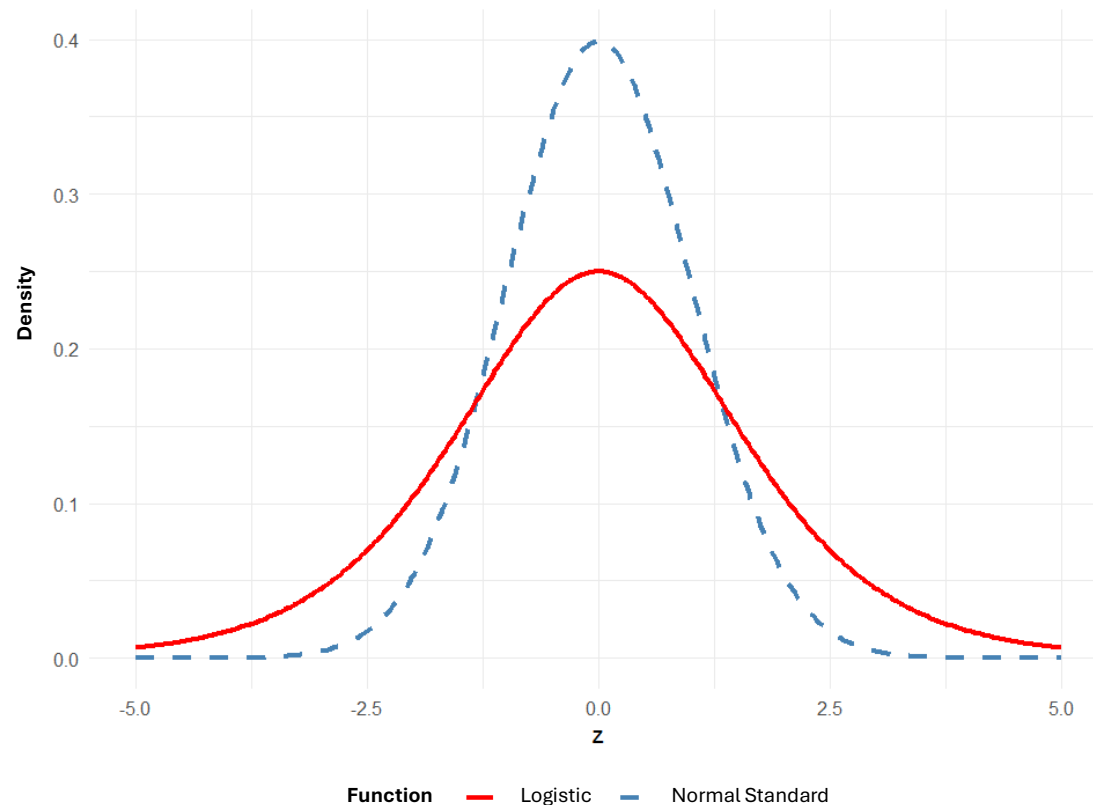
$$\Lambda(z) = \frac{e^z}{1 + e^z}$$



Binomial Models

Logit and Probit models

Density Functions: Normal vs Logistic



Key Differences Between the Normal and Logistic Densities

- The logistic distribution has a larger variance, which means its density is more spread out.
- In practice, it assigns more probability to values farther from the centre. Because of that, the logistic density has heavier tails than the standard normal density.

Because the logistic density has heavier tails, **it assigns relatively more probability to extreme values of the predictor z** compared with the probit model.

Binomial Models

Logit and Probit models

Given:

$$\mu_n = F(\eta_n), \text{ con: } \eta_n = \alpha + \beta^\top x_n$$

the k -th covariate changes, its marginal effect on the probability is given by:

$$\frac{\partial \mu_n}{\partial x_{nk}} = f(\eta_n) \cdot \beta_k$$

Where f is the first derivative of the link function F

- ϕ : The standard normal density in the probit model, .
- λ : The logistic density in the logit model.

Thus, the marginal effect is obtained by multiplying the coefficient β_k by $f(\eta_n)$, the value of the density function evaluated at each observation's linear predictor.

This implies that marginal effects vary across observations, but:

- the ratio of two marginal effects for any pair of covariates does not depend on the observation, because it equals the ratio of their coefficients.

If we compare two covariates $x_{\{nk\}}$ and $x_{\{nj\}}$, their marginal effects are:

$$\frac{\partial \mu_n}{\partial x_{nk}} = f(\eta_n) \cdot \beta_k \quad \text{y} \quad \frac{\partial \mu_n}{\partial x_{nj}} = f(\eta_n) \cdot \beta_j$$

Then The ratio is independent of the specific values of the covariates for observation n :

$$\frac{\frac{\partial \mu_n}{\partial x_{nk}}}{\frac{\partial \mu_n}{\partial x_{nj}}} = \frac{f(\eta_n) \cdot \beta_k}{f(\eta_n) \cdot \beta_j} = \frac{\beta_k}{\beta_j}$$

Binomial Models

Logit and Probit models

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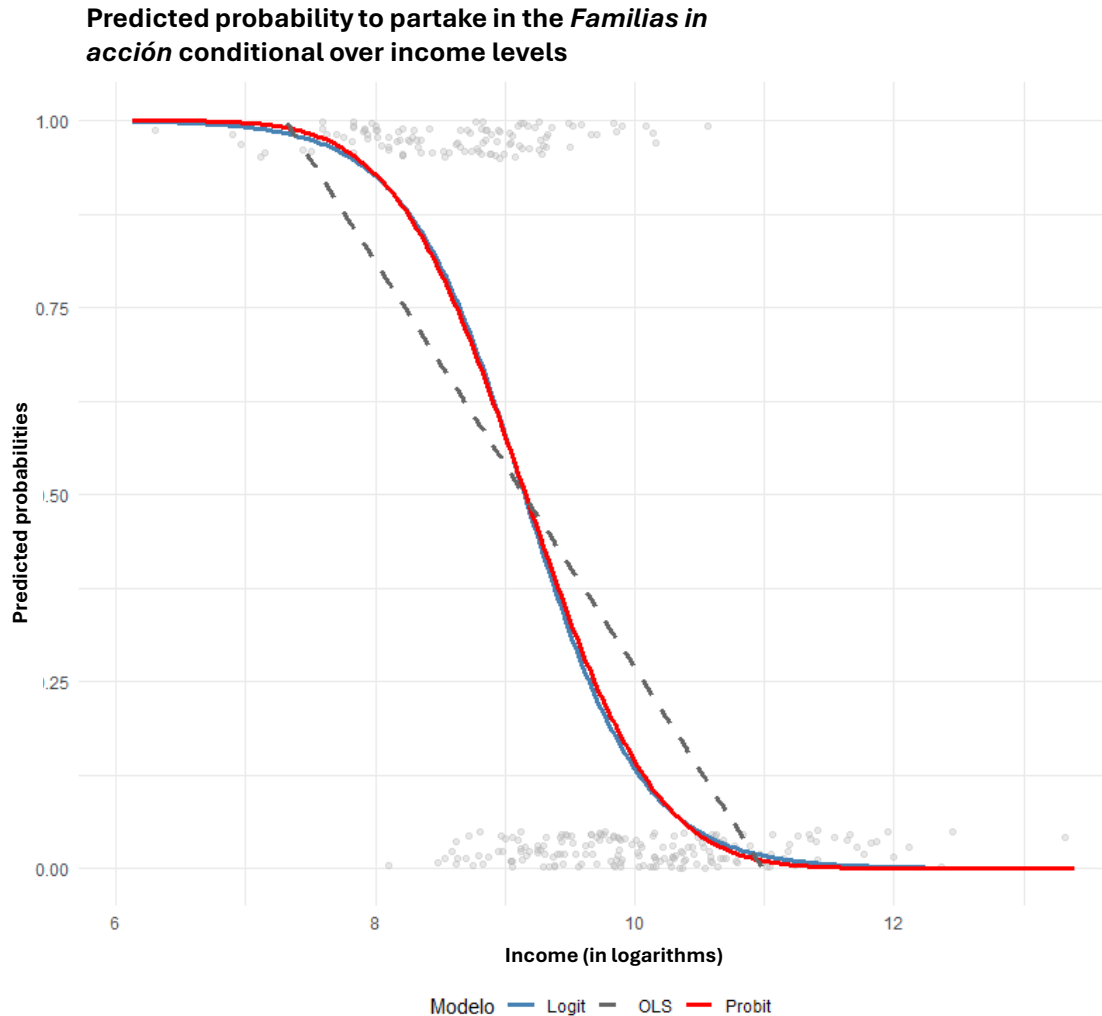
Because the density functions act as proportionality factors, the maximum marginal effect occurs when $\eta_n = 0$, which corresponds to a success probability of 0.5

At that point, the values of the density functions are:

- $\phi(0) \approx 0.4$
- $\lambda(0) \approx 0.25$

Therefore, a useful rule of thumb for interpreting the coefficients is to multiply them by 0.4 in the probit model and by 0.25 in the logit model to obtain an estimate of the maximum marginal effect.

Binomial Models



Logit

$$p_i = \Pr(y_i = 1 \mid x_i) = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Probit

$$p_i = \Pr(y_i = 1 \mid x_i) = \phi(\beta_0 + \beta_1 x_i)$$

Linear probability

$$p_i = \Pr(y_i = 1 \mid x_i) = \beta_0 + \beta_1 x_i$$

Common models to study binary outcome variables

Model	Probability $p = \Pr[y = 1 \mid \mathbf{x}]$	Marginal Effect $\frac{\partial p}{\partial x_j}$
Logit	$\Lambda(\mathbf{x}'\boldsymbol{\beta}) = \frac{e^{\mathbf{x}'\boldsymbol{\beta}}}{1 + e^{\mathbf{x}'\boldsymbol{\beta}}}$	$\Lambda(\mathbf{x}'\boldsymbol{\beta})[1 - \Lambda(\mathbf{x}'\boldsymbol{\beta})]\boldsymbol{\beta}_j$
Probit	$\Phi(\mathbf{x}'\boldsymbol{\beta}) = \int_{-\infty}^{\mathbf{x}'\boldsymbol{\beta}} \phi(z) dz$	$\phi(\mathbf{x}'\boldsymbol{\beta})\boldsymbol{\beta}_j$
Lineal Probability	$\mathbf{x}'\boldsymbol{\beta}$	$\boldsymbol{\beta}_j$

Binomial models

The Logit Model

The Logit Model

The logit model specification is:

$$p = \Lambda(\mathbf{x}'\beta) = \frac{e^{\mathbf{x}'\beta}}{1 + e^{\mathbf{x}'\beta}}$$

Shifting the numerator, to a denominator product, we get:

$$p = \frac{1}{(1 + e^{\mathbf{x}'\beta})(e^{\mathbf{x}'\beta})^{-1}}$$

The product in the denominator turns into:

$$p = \frac{1}{e^{-\mathbf{x}'\beta} + e^{\mathbf{x}'\beta - \mathbf{x}'\beta}}$$

Adding the exponent of common bases:

$$p = \frac{1}{1 + e^{-\mathbf{x}'\beta}}$$

Reordering

$$p + pe^{-\mathbf{x}'\beta} = 1$$

$$\frac{1 - p}{p} = e^{-\mathbf{x}'\beta}$$

If we apply log to the expression, we get:

$$\ln\left(\frac{1 - p}{p}\right) = \ln(e^{-\mathbf{x}'\beta})$$

Using the fact that the fraction of a log is the difference of its term, we get:

$$\ln(1 - p) - \ln(p) = -\mathbf{x}'\beta$$

$$\ln(p) - \ln(1 - p) = \mathbf{x}'\beta$$

$$\ln\left(\frac{p}{1 - p}\right) = \mathbf{x}'\beta$$

The probability that $y = 1$ regarding the probability that $y = 0$ is called the **Odds Ratio** or **Relative Risk Ratio**.

The Logit Model

From

Probability → to odds → to log odds

Suppose that the probability of success of an event is 0.8. Then the probability of failure is: $1 - 0.8 = 0.2$

The odds of success are defined as the ratio between the probability of success and the probability of failure:

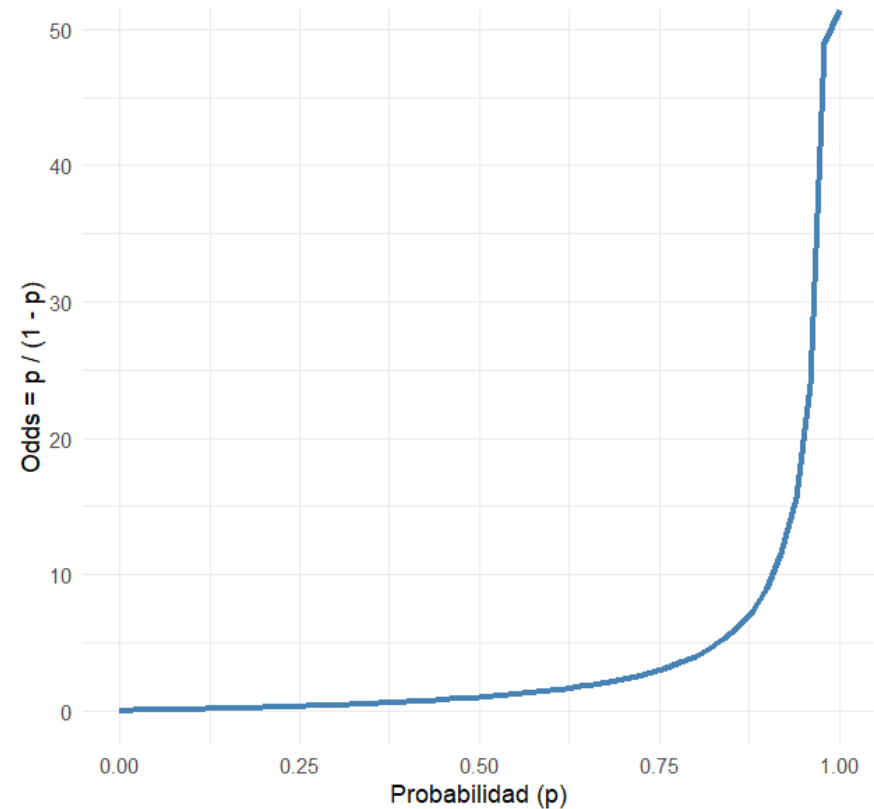
$$\text{Odds} = \frac{p}{1 - p}$$

Probabilidad (p)	Odds
0.001	0.0010
0.010	0.0101
0.150	0.1765
0.200	0.2500
0.250	0.3333
0.300	0.4286
0.350	0.5385
0.400	0.6667
0.450	0.8182
0.500	1.0000
0.550	1.2222
0.600	1.5000
0.650	1.8571
0.700	2.3333
0.750	3.0000
0.800	4.0000
0.850	5.6667
0.900	9.0000
0.999	999.0000
0.9999	9999.0000

The transformation from probability to odds is monotonically increasing: **as the probability increases, the odds also increase.**

Additionally, probability is bounded between 0 and 1, whereas odds range from 0 to positive infinity.

Odds and Probability



The Logit Model

From

Probability → to odds → to log odds

The transformation from odds to log-odds (logit) is performed using the natural logarithm:

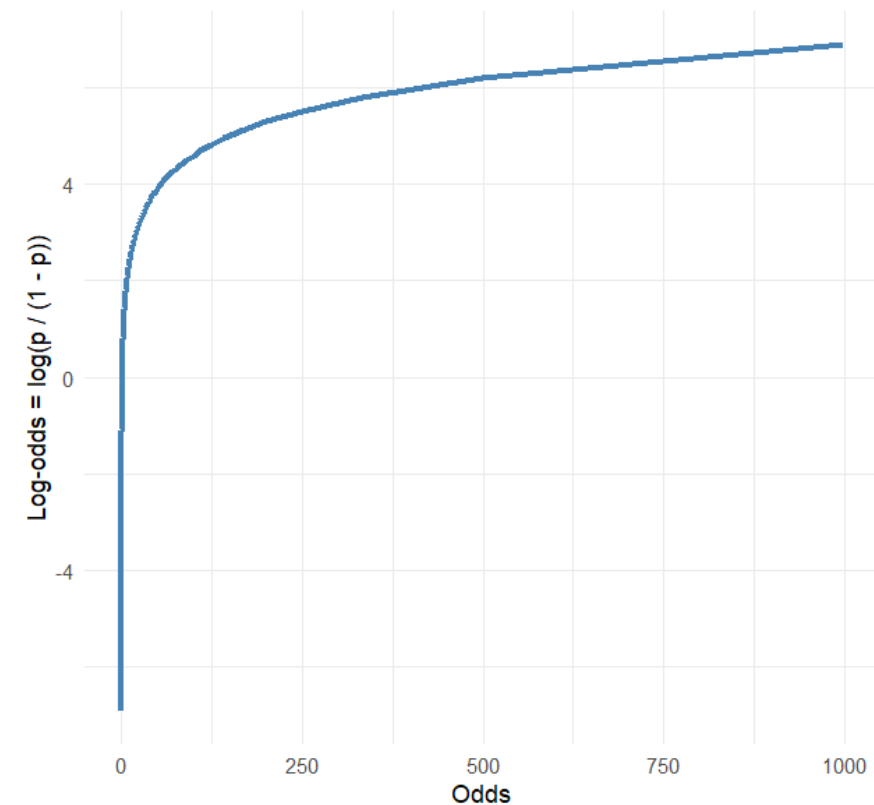
$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

Probabilidad (p)	Odds	Log-odds
0.001	0.0010	-6.9068
0.010	0.0101	-4.5951
0.150	0.1765	-1.7346
0.200	0.2500	-1.3863
0.250	0.3333	-1.0986
0.300	0.4286	-0.8473
0.350	0.5385	-0.6190
0.400	0.6667	-0.4055
0.450	0.8182	-0.2007
0.500	1.0000	0.0000
0.550	1.2222	0.2007
0.600	1.5000	0.4055
0.650	1.8571	0.6190
0.700	2.3333	0.8473
0.750	3.0000	1.0986
0.800	4.0000	1.3863
0.850	5.6667	1.7346
0.900	9.0000	2.1972
0.999	999.0000	6.9068
0.9999	9999.0000	9.2102

This transformation is also monotonically increasing: higher odds correspond to higher log-odds.

Its usefulness lies in the fact that it converts a bounded variable (probability between 0 and 1) into an unbounded one (log-odds range from $-\infty$ to $+\infty$), which simplifies statistical modelling.

Odds and Log Odds



The Logit Model

$$\ln\left(\frac{p}{1-p}\right) = \mathbf{x}'\boldsymbol{\beta}$$

Consider a pharma study where: $y = 1$ indicates **Survival** $y = 0$ **death**, and regressors include a measurement of medication consumption

$$\text{State of paciente}(y) = \begin{cases} 1 & \text{si } y = \text{Survival} \\ 0 & \text{si } y = \text{Death} \end{cases}$$

The Odds Ratio is defined as:

$$\text{Odds} = \frac{\Pr(y = 1 \mid \mathbf{x})}{\Pr(y = 0 \mid \mathbf{x})} = \frac{p}{1-p}$$

In the logit model the odd ratio is linear in the regressors:

$$e^{\left(\ln\left(\frac{p}{1-p}\right)\right)} = e^{(\mathbf{x}'\boldsymbol{\beta})} = \frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})}$$

$\mathbf{x}'\boldsymbol{\beta}$ is the lineal index of the model, which describes the scalar product between:

- $\mathbf{x} = (x_1, x_2, \dots, x_k)$: vector of **regressors**.
- $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_k)$: vector of **coefficients**.

Then:

$$\mathbf{x}'\boldsymbol{\beta} = x_1\beta_1 + x_2\beta_2 + \dots + x_j\beta_j + \dots + x_k\beta_k$$

Suppose the **j-th regressor increases in and unit**, that is x_j moves from x_j to $x_j + 1$. Then, the new lineal index is:

$$x_1\beta_1 + x_2\beta_2 + \dots + (x_j + 1)\beta_j + \dots + x_k\beta_k = \mathbf{x}'\boldsymbol{\beta} + \beta_j$$

Then, if the index shifts from $\mathbf{x}'\boldsymbol{\beta}$ to $\mathbf{x}'\boldsymbol{\beta} + \beta_j$, the probability ratio y multiplied by $e^{(\beta_j)}$:

$$\frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})} \longrightarrow e^{(\mathbf{x}'\boldsymbol{\beta} + \beta_j)} = \underbrace{e^{(\mathbf{x}'\boldsymbol{\beta})}}_{\text{Odds}} \cdot e^{(\beta_j)}$$

The Odds ratios increases by a factor of $e^{(\beta_j)}$

The Logit Model

Therefore, a parameter of 0.1 means that an increase of one unit in the regressor multiplies the initial odds ratio by:

$$e^{(0.1)} \approx 1.105$$

This means that the odds ratio increases by 1.105, that is, the relative probability of survival increases by 10.5%:

$$e^{(0.1)} - 1 = 1.105 - 1 = 0.105$$

Although the odds increase by 10.5%, the probability p of survival does not increase by that same percentage, **because the relationship between odds and probability is non-linear:**

$$p = \frac{\text{Odds}}{1 + \text{Odds}}$$

One unit change impact over (x_j) with ($\beta_j = 0.1$)

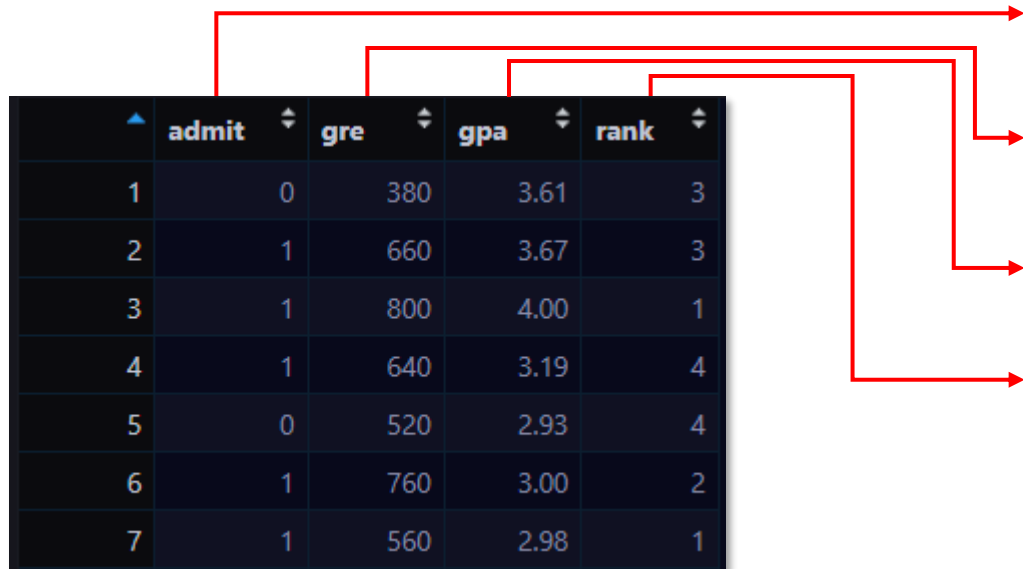
$z = \mathbf{x}'\beta$	p	Odds	$z + 0.1$	Nueva Odds	p_{nuevo}
-2	0.119	0.135	-1.9	0.149	0.130
-1	0.269	0.368	-0.9	0.406	0.289
0	0.500	1.000	0.1	1.105	0.524
1	0.731	2.718	1.1	3.004	0.750
2	0.881	7.389	2.1	8.165	0.891

- When $z = 0$, the probability is 0.5 and the Odds ratio is 1. An unit increase in x_j with $\beta_j = 0.1$ leads to an OR of 1.105 y and to a 0.524 probability.
- The effect on the probability is not constant: it depends on the starting point of z .
- The multiplier $e^{(\beta_j)}$ always applies to the odds, but its effect on p varies depending on the context.

The Logit Model

A researcher is interested in how variables such as GRE score (Graduate Record Examination), GPA (Grade Point Average), and the prestige of the undergraduate institution **influence admission to postgraduate programmes.**

The response variable, admitted/not admitted, is binary.



	admit	gre	gpa	rank
1	0	380	3.61	3
2	1	660	3.67	3
3	1	800	4.00	1
4	1	640	3.19	4
5	0	520	2.93	4
6	1	760	3.00	2
7	1	560	2.98	1

- **Admit.** Dichotomous dependent variable indicating whether the person was admitted to the postgraduate programme.
- **GRE.** Independent variable, integer numerical, describing the score obtained in the Graduate Record Examination.
- **GPA.** Independent variable, discrete numerical variable describing the student's cumulative academic average (Grade Point Average).
- **Rank.** Ordinal variable describing the ranking of the university from which each student graduated.

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7	1	560	2.98	1

$$\Pr(Y = 1 \mid x_1 \dots x_k) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)}}$$

$$\Pr(y_1 = 1 \mid gre_1, gpa_1, rank_1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 gre_1 + \beta_2 gpa_2 + \beta_3 rank_2)}}$$

The Logit Model

Before fitting the model, let us begin with an EDA using cross-tabulations and sample statistics:

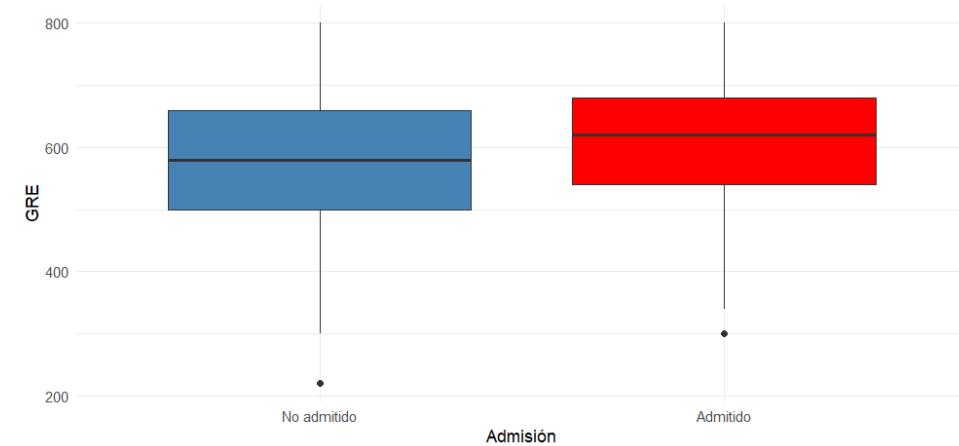
Descriptive statistics of the sample

Statistics	admit	gre	gpa	rank
Min	0.0000	220.0	2.260	1.000
1st quartile	0.0000	520.0	3.130	2.000
Median	0.0000	580.0	3.395	2.000
Mean	0.3175	587.7	3.390	2.485
3th quartile	1.0000	660.0	3.670	3.000
Max	1.0000	800.0	4.000	4.000

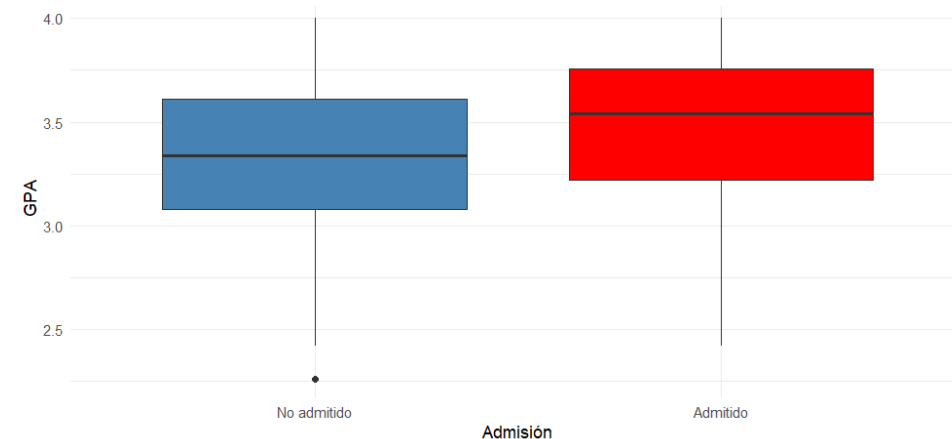
Cross-tabulation between the dependent variable and university rank

Admitted	Rank 1	Rank 2	Rank 3	Rank 4
No	28	97	93	55
Sí	33	54	28	12

Distribution of GRE scores by admission



Distribution of GPA scores by admission



The Logit Model

When estimating the model, we obtain:

Generalised regression function

Reference to the table that contains the data records

Dependent variable

Independent variables

Link function

Estimated coefficients of the logit model. Their units are log-odds and cannot be interpreted in the same way as OLS coefficients. To interpret them, one must:

- compute odds ratios
- evaluate predicted probabilities.

Two-tailed p-values, which evaluate the alternative hypothesis that each coefficient is different from zero.

- To reject the null hypothesis associated with a coefficient, the p-value must be smaller than α (usually 0.05).

Statistic for testing whether each coefficient is different from zero. To reject the null hypothesis, the value of the statistic must be greater than the critical value associated with α (1.96 if the significance level is 0.05).

Deviance of the null model (without predictors): indicates how well (or poorly) the model predicts the response variable without using the controls.

Degrees of freedom of the model: equal to the number of observations minus the number of estimated parameters.

Given 400 observations: the null model (no predictors) has 399 degrees of freedom, the model with controls has 396 degrees of freedom (400 minus the 4 parameters estimated).

AIC (Akaike Information Criterion): penalizes model complexity; lower AIC indicates a better balance between fit and parsimony.

Deviance of the model with predictors

```
Call:
glm(formula = admit ~ gre + gpa + rank, family = "binomial",
    data = dt)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.989979   1.139951  -3.500 0.000465 ***
gre           0.002264   0.001094   2.070 0.038465 *
gpa           0.804038   0.331819   2.423 0.015388 *
rank2        -0.675443   0.316490  -2.134 0.032829 *
rank3        -1.340204   0.345306  -3.881 0.000104 ***
rank4        -1.551464   0.417832  -3.713 0.000205 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 499.98  on 399  degrees of freedom
Residual deviance: 458.52  on 394  degrees of freedom
AIC: 470.52

Number of Fisher Scoring iterations: 4
```

The Logit Model

Results of the logistic regression regarding the probability of acceptance

Variable	Coeff	Standard Error	z	P-value p
Intercept	-3.990	1.140	-3.500	$p < 0.001^{***}$
GRE	0.002	0.001	2.070	$p = 0.038^*$
GPA	0.804	0.332	2.423	$p = 0.015^*$
Rank (2)	-0.675	0.316	-2.134	$p = 0.033^*$
Rank (3)	-1.340	0.345	-3.881	$p < 0.001^{***}$
Rank (4)	-1.551	0.418	-3.713	$p < 0.001^{***}$

Significance levels: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$
 Sample size: 400 observations
 Null deviations: 499.98 (gl = 399)
 Residual deviations: 458.52 (gl = 394)
 AIC: 470.52
 Fisher scoring iterations: 4
 Dispersion: 1 (Binomial model)

- For each additional unit in the GRE score, **the log-odds of admission (versus non-admission) increase by 0.002.**
- For each additional unit in GPA, **the log-odds of being admitted to postgraduate studies increase by 0.804.**
- The indicator variables for institutional prestige have a slightly different interpretation.**
 - For example, graduating from a Rank 2 institution, compared with a Rank 1 institution, reduces the log-odds of admission by -0.675.

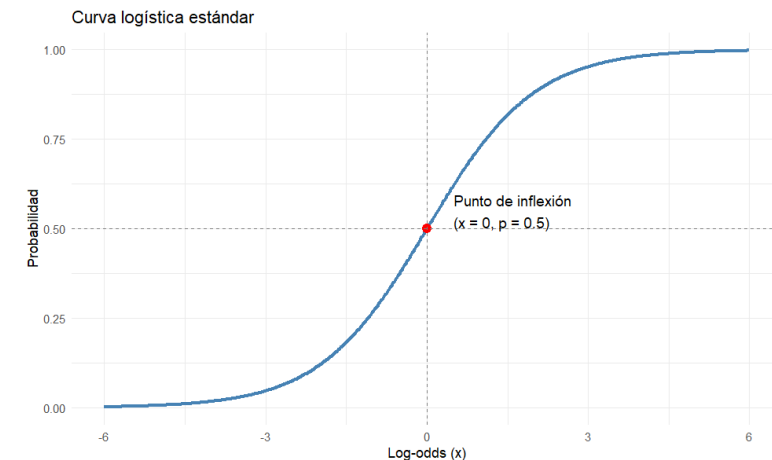
The derivative of the logistic function $f(x) = \frac{1}{1+e^{-x}}$ reaches its maximum at $x = 0$, where:

$$f'(0) = \frac{e^0}{(1+e^0)^2} = \frac{1}{(1+1)^2} = \frac{1}{4} = 0.25$$

This value represents the maximum change in probability resulting from a one-unit change in the log-odds, **occurring exactly at the inflection point of the logistic curve.**

When the model is near the centre of the distribution (probability ≈ 0.5), the change in probability may be approximated as:

$$\Delta \text{probability} \approx \beta \times 0.25$$



The Logit Model

Results of the logistic regression regarding the probability of acceptance

Variable	Coeff	Standard Error	z	P-value p
Intercept	-3.990	1.140	-3.500	$p < 0.001^{***}$
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This value represents the maximum change in probability resulting from a one-unit change in the log-odds, **occurring exactly at the inflection point of the logistic curve.**

$$\Delta \text{probability} \approx \beta \times 0.25$$

- The marginal effect of GRE is small:** each additional GRE point increases the probability of admission by about 0.05% when the baseline probability is around 0.5
- GPA has a large effect:** a one-unit increase in GPA raises the probability of admission by approximately 20 percentage points.
- Institutional prestige reduces admission chances:** moving from Rank 1 to Rank 4 decreases the probability of admission by nearly 39 percentage points, holding the other covariates constant.

The Logit Model

To interpret the results, we can (and should) also apply the **exponential function to the estimated** parameters to obtain the odds ratios (ORs):

$$\frac{p}{1-p} = e^{(\mathbf{x}'\beta)}$$

However, we are not only interested in the exponents themselves, **but in the multiplicative effect on the odds of a one-unit change in the predictor** x_j , holding all other variables constant.

$$\frac{p}{1-p} = e^{(\mathbf{x}'\beta)} \longrightarrow e^{(\mathbf{x}'\beta + \beta_j)} = \underbrace{e^{(\mathbf{x}'\beta)}}_{\text{Odds}} \cdot e^{(\beta_j)}$$

Results of the logistic regression regarding the probability of acceptance in terms of odds ratios (OR)

Variable	OR	Lower limit	Upper limit
Intercept	0.0185	0.0019	0.1665
GRE	1.0023	1.0001	1.0044
GPA	2.2345	1.1739	4.3238
Rank (2)	0.5089	0.2723	0.9448
Rank (3)	0.2618	0.1316	0.5115
Rank (4)	0.2119	0.0907	0.4707

GRE

1. Odds ratio

The estimated coefficient for GRE is $\beta = 0.0023$, meaning that for each **additional point in the GRE score, the log-odds of being admitted increase by 0.0023 units**. This translates into an **odds ratio of:**

$$e^{0.0023} \approx 1.0023$$

$$(1.0023 - 1) \times 100 = 0.23\%$$

That is, a 0.23% increase in the odds of admission for every additional GRE point.

2. 0.25 rule

To get the maximum change in probability resulting from a one-unit change in the log-odds, **we use the inflection point of the logistic curve**.

$$\Delta p \approx 0.0023 \times 0.25 = 0.000575$$

$$\approx 0.0023 \times 0.25 \times 10 = 0.575$$

This corresponds to an increase of about 0.0575 percentage points in the probability of admission per additional **10 GRE points**, assuming **the baseline probability is near 0.5**.

The Logit Model

To interpret the results, we can (and should) also apply the **exponential function to the estimated** parameters to obtain the odds ratios (ORs):

$$\frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})}$$

However, we are not only interested in the exponents themselves, **but in the multiplicative effect on the odds of a one-unit change in the predictor** x_j , holding all other variables constant.

$$\frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})} \longrightarrow e^{(\mathbf{x}'\boldsymbol{\beta} + \beta_j)} = \underbrace{e^{(\mathbf{x}'\boldsymbol{\beta})}}_{\text{Odds}} \cdot e^{(\beta_j)}$$

Results of the logistic regression regarding the probability of acceptance in terms of odds ratios (OR)

Variable	OR	Lower limit	Upper limit
Intercept	0.0185	0.0019	0.1665
GRE	1.0023	1.0001	1.0044
GPA	2.2345	1.1739	4.3238
Rank (2)	0.5089	0.2723	0.9448
Rank (3)	0.2618	0.1316	0.5115
Rank (4)	0.2119	0.0907	0.4707

GPA

1. Odds ratio

The estimated coefficient for GPA is $\beta = 0.084$, meaning that for each **additional point in the GRE score, the log-odds of being admitted increase by 0.084 units**. This translates into an **odds ratio of:**

$$e^{0.084} \approx 2.2345$$

$$(2.2345 - 1) \times 100 = 123.45\%$$

That is, a 123.45% increase in the odds of admission for every additional GRE point.

2. 0.25 rule

To get the maximum change in probability resulting from a one-unit change in the log-odds, **we use the inflection point of the logistic curve**.

$$\Delta p \approx 0.804 \times 0.25 = 0.201$$

$$\approx 0.0023 \times 0.25 \times 10 = 201$$

This corresponds to an increase of about 20 percentage points in the probability of admission per additional **10 GPA points, assuming the baseline probability is near 0.5**.

The Logit Model

To interpret the results, we can (and should) also apply the **exponential function to the estimated** parameters to obtain the odds ratios (ORs):

$$\frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})}$$

However, we are not only interested in the exponents themselves, **but in the multiplicative effect on the odds of a one-unit change in the predictor** x_j , holding all other variables constant.

$$\frac{p}{1-p} = e^{(\mathbf{x}'\boldsymbol{\beta})} \longrightarrow e^{(\mathbf{x}'\boldsymbol{\beta} + \beta_j)} = \underbrace{e^{(\mathbf{x}'\boldsymbol{\beta})}}_{\text{Odds Ratio}} \cdot e^{\beta_j}$$

Results of the logistic regression regarding the probability of acceptance in terms of odds ratios (OR)

Variable	OR	Lower limit	Upper limit
Intercept	0.0185	0.0019	0.1665
GRE	1.0023	1.0001	1.0044
GPA	2.2345	1.1739	4.3238
Rank (2)	0.5089	0.2723	0.9448
Rank (3)	0.2618	0.1316	0.5115
Rank (4)	0.2119	0.0907	0.4707

Rank 4 vs 1

1. Odds ratio

The estimated coefficient for Rank 4 is $\beta = -1.552$, which implies that having completed undergraduate studies at a Rank 4 institution, compared with a Rank 1 institution, reduces the log-odds of admission by 1.552 units. This translates into an **odds ratio of:**

$$e^{-1.552} \approx 0.2119$$

$$(1 - 0.2119) \times 100 = 78.818\%$$

The odds of **admission decrease by about 78.81%** when moving from a Rank 1 to a Rank 4 institution, holding all other variables constant.

A student graduation from a Rank 4 university has an odds of acceptance of 0.2119, relative to that of graduates from Rank 1 university.

$$\frac{1}{0.2119} \approx 4.72$$

For every 10 expected admissions for Rank 4 graduates, we would expect up to 47 admissions for Rank 1 graduates, keeping all other things constant.

Applying the 0.25 rule:

$$\Delta p \approx -1.552 \times 0.25 = -0.388$$

That's a estimated decrease of 38.8 percentual points in the odds of admission for graduates from Rank 4 universities, relative to graduates from Rank 1 universities, **if the base probability is close to 0.5.**

The Logit Model

Build population profiles and to understand changes in probability based on average values of the control variables.

DV values and predicted probabilities of admission

GRE	GPA	Rank	RankP
587.7	3.3899	1	0.5166
587.7	3.3899	2	0.3523
587.7	3.3899	3	0.2186
587.7	3.3899	4	0.1847

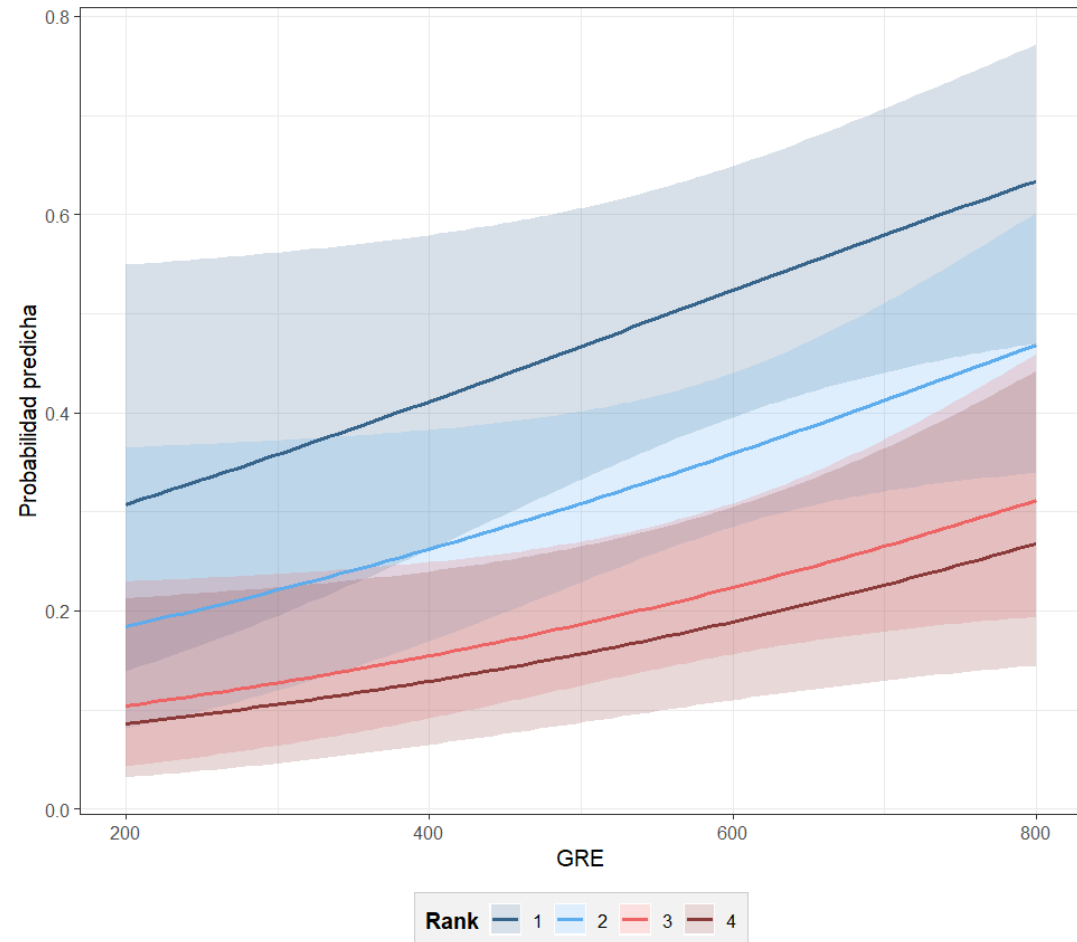
$$\Pr(y_1 = 1 \mid X) = \frac{1}{1 + e^{-((-3.989979 + (0.002264 \times 587.7) + (0.804038 \times 3.3899) + (0)))}} = 0.5165395$$

Logit fit

Variable	Coeff	Standard Error	Z statistic	P-value
Intercept	-3.990	1.140	-3.500	$p < 0.001^{***}$
GRE	0.002	0.001	2.070	$p = 0.038^*$
GPA	0.804	0.332	2.423	$p = 0.015^*$
Rank (2)	-0.675	0.316	-2.134	$p = 0.033^*$
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Rank (4)	-1.551	0.418	-3.713	$p < 0.001^{***}$

The logit model

Predicted probability of being admitted to a postgraduate programme, given **University prestige levels, and GRE score**



The logistic model not only allows us to estimate individual probabilities, but also to construct comparative profiles of admission. By keeping GPA constant and systematically varying GRE and Rank, we can identify concrete cases of admission probability.



María

Rank: low
GRE: higher than the average mean 800
GPA: higher than the average mean 4

Admission probability : 37% (SE. 0.094)



Juan David

Rank: high
GRE: close to the average mean 600
GPA: lower than the average mean 3

Admission probability : 44% (SE. 0.074)